

Backtracking (Assignment Solutions)

Solution 1:

Hint : To track which cell has or not been visited, create a NxN vector called visited.

This vector will be initialized with false values for all cells & make the value for a particular cell to true when you have visited it.

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

void solveMazeUtil(int maze[][4], int x, int y, string sol, int N,
vector<vector<bool>> vis) {
    if (x == N - 1 && y == N - 1 && maze[x][y] == 1) {
        cout << sol << "\n";
        return;
    }

    //up
    if(x-1 >= 0 && !vis[x-1][y] && maze[x-1][y] == 1) {
        vis[x][y] = true;
        solveMazeUtil(maze, x-1, y, sol+"U", N, vis);
        vis[x][y] = false;
    }

    //down
    if(x+1 < N && !vis[x+1][y] && maze[x+1][y] == 1) {
        vis[x][y] = true;
        solveMazeUtil(maze, x+1, y, sol+"D", N, vis);
        vis[x][y] = false;
    }

    //right
    if(y+1 < N && !vis[x][y+1] && maze[x][y+1] == 1) {
        vis[x][y] = true;
        solveMazeUtil(maze, x, y+1, sol+"R", N, vis);
    }
}
```

```
        vis[x][y] = false;
    }

    //left
    if(y-1 >= 0 && !vis[x][y-1] && maze[x][y-1] == 1) {
        vis[x][y] = true;
        solveMazeUtil(maze, x, y-1, sol+"L", N, vis);
        vis[x][y] = false;
    }
}

void solveMaze(int maze[][4], int N) {
    string sol = "";
    vector<vector<bool>> vis(N, vector<bool> (N, false));

    if(maze[0][0] == 1) { // only move if the initial position allows
        solveMazeUtil(maze, 0, 0, sol, N, vis);
    }
}

int main(){
    int n = 4;
    int maze[4][4] = { { 1, 0, 0, 0 },
                        { 1, 1, 0, 1 },
                        { 1, 1, 0, 0 },
                        { 0, 1, 1, 1 } };

    solveMaze(maze, n);
}
```

Solution 2:

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

void bfs(int pos, int len, string ans, string Dig, vector<vector<char>> L) {
    if (pos == len){
        cout << ans << endl;
    } else {
        vector<char> letters = L[Dig[pos]-'0'];
        for (int i = 0; i < letters.size(); i++){
            bfs(pos+1, len, ans + letters[i], Dig, L);
        }
    }
}

void letterCombinations(string Dig, vector<vector<char>> L) {
    int len = Dig.size();
    if (len == 0) {
        cout << "";
        return;
    }

    string ans = "";
    bfs(0, len, ans, Dig, L);
}

int main(){
    vector<vector<char>> L = {{},{},{ 'a','b','c'},{ 'd','e','f'},{ 'g','h','i' },
                             { 'j','k','l'},{ 'm','n','o'},{ 'p','q','r','s' },
                             { 't','u','v'},{ 'w','x','y','z' }};

    string digits = "23";
    letterCombinations(digits, L);
}
```

Solution 3 :

```
#include <iostream>
#include <string>
#include <vector>
using namespace std;

bool isSafe(int x, int y, int sol[][8], int N){
    return (x >= 0 && x < N && y >= 0 && y < N
            && sol[x][y] == -1);
}

void printSolution(int sol[][8], int N) {
    for (int x = 0; x < N; x++) {
        for (int y = 0; y < N; y++)
            cout << sol[x][y] << " ";
        cout << endl;
    }
}

bool solveKTUtil(int x, int y, int movei, int sol[][8],
                 int xMove[],int yMove[], int N) {

    int k, next_x, next_y;
    if (movei == N * N)
        return true;

    for (k = 0; k < 8; k++) {
        next_x = x + xMove[k];
        next_y = y + yMove[k];
        if (isSafe(next_x, next_y, sol, N)) {
            sol[next_x][next_y] = movei;
            if (solveKTUtil(next_x, next_y, movei + 1,
                           sol, xMove, yMove, N))
                return true;
            else
                sol[next_x][next_y]
                    = -1; // backtracking
        }
    }
}
```

```
        return false;
    }

bool solveKT(int N) {
    int sol[8][8];
    for (int x = 0; x < N; x++)
        for (int y = 0; y < N; y++)
            sol[x][y] = -1;

    int xMove[] = { 2, 1, -1, -2, -2, -1, 1, 2 };
    int yMove[] = { 1, 2, 2, 1, -1, -2, -2, -1 };

    //As the Knight starts from cell(0,0)
    sol[0][0] = 0;

    if (!solveKTUtil(0, 0, 1, sol, xMove, yMove, N)) {
        cout << "Solution does not exist\n";
        return false;
    }
    else
        printSolution(sol, N);

    return true;
}

int main(){
    int N = 8;
    solveKT(N);
}
```